Java Process Table Function (PTF) User-Defined Function (UDF) type — User Event Enricher, a state-driven PTF example

> The User Event Enricher is driven entirely by state transitions triggered by incoming rows.
>
> Rather than a row-at-a-time transformation, it behaves as a stateful operator that maintains and evolves per-user state across events.
>
> This example demonstrates how the Process Table Function (PTF) API in Flink 2.1+ enables building fully stateful operators in Java that are directly callable from SQL.

Table of Contents

- **1.0 State and operators in a Process Table Function**
  - **1.1 What is state?**
  - **1.2 How operators use state**
  - **1.3 How `PARTITION BY` connects SQL to state**
  - **1.4 The role of `@StateHint` and `@ArgumentHint`**
- **2.0 What does this example do?**
  - **2.1 Enrichment logic**
  - **2.2 How it works end-to-end**
  - **2.3 Key concepts illustrated**
- **3.0 Resources**

1.0 State and operators in a Process Table Function

A **Process Table Function (PTF)** is a new category of user-defined function introduced in Flink 2.1. Unlike scalar or aggregate UDFs, a PTF is a **stateful operator** — it sits inside the Flink dataflow graph just like a built-in operator (e.g., a windowed aggregation or a keyed process function), but you define its logic in a plain Java class.

1.1 What is state?

In stream processing, **state** is data that persists across events. Without state every event is processed in isolation — you can filter or transform it, but you cannot count things, detect sequences, or remember what happened before. State is what turns a stateless pipe into an intelligent processor.

Flink manages state for you: it stores it in a **state backend** (heap, RocksDB, etc.), checkpoints it for fault tolerance, and restores it on failure. You never serialize or recover state manually.

1.2 How operators use state

Every Flink operator that needs memory between events holds state. A few examples:

Operator	State it keeps

Operator	State it keeps
Keyed window aggregation	Partial aggregates per key per window
Interval join	Buffered rows from both sides within the join window
Deduplication	Set of seen keys
ProcessTableFunction (PTF)	Whatever you declare via `@StateHint` — one instance per partition key

The PTF is special because **you** decide what the state looks like. You declare a POJO, annotate it with `@StateHint`, and Flink handles the rest — creating one instance per partition key, persisting it in the state backend, and checkpointing it automatically.

1.3 How `PARTITION BY` connects SQL to state

When you write:

```
SELECT *
FROM TABLE(
    UserEventEnricher(
        input => TABLE user_events PARTITION BY user_id
    )
)
```

Flink does the following behind the scenes:

1. **Key the stream** — `PARTITION BY user_id` tells the Flink planner to hash-partition the input by `user_id`, exactly like a `keyBy()` in the DataStream API.
2. **Allocate isolated state** — for each distinct `user_id`, Flink creates a separate `UserState` POJO instance. For example, as shown in the sample data, Alice's state never overlaps with Bob's.
3. **Route events** — every incoming row is routed to the partition (and therefore the state instance) that matches its `user_id`.
4. **Call `eval()`** — your `eval()` method receives the row *and* the correct state instance already loaded. You read and mutate the POJO fields directly — no `ValueState.get()` / `.update()` boilerplate.
5. **Checkpoint** — Flink snapshots all state instances periodically. On failure it restores them and replays from the last checkpoint, so your PTF is exactly-once by default.

1.4 The role of `@StateHint` and `@ArgumentHint`

These two annotations are the bridge between your Java code and the Flink operator graph:

- `@StateHint` — marks a parameter as operator state. Flink sees the annotated POJO and wires it into the keyed state backend. Each `PARTITION BY` key gets its own instance, automatically serialized, checkpointed, and restored.
- `@ArgumentHint(ArgumentTrait.SET_SEMANTIC_TABLE)` — declares that the input argument is a *set-semantic table*, meaning the PTF acts as a keyed, stateful virtual processor over the entire partitioned stream. This is what distinguishes a PTF from a simple row-at-a-time scalar UDF.

Together, these annotations let you write what *looks* like a plain method but *executes* as a fully fault-tolerant, distributed, keyed-state operator inside the Flink pipeline.

2.0 What does this example do?

This example puts the above concepts into practice. The `UserEventEnricher` PTF reads raw user-interaction events from a Kafka topic (`user_events`), enriches each event with session and counting information, and writes the result to a second Kafka topic (`enriched_events`).

2.1 Enrichment logic

For every incoming event the function maintains three pieces of **per-user state** (one state instance per `PARTITION BY user_id` key):

State field	Purpose
`sessionId`	Monotonically increasing session counter. Incremented each time a `"login"` event arrives.
`eventCount`	Running count of events **within the current session**. Reset to zero on each new session, then incremented for every event (including the login itself).
`lastEvent`	The `event_type` of the most recent event for that user.

Each incoming row is emitted immediately with these three extra fields appended, so the output schema is:

```
user_id     STRING    — passed through automatically via PARTITION BY
event_type  STRING
payload     STRING
session_id  BIGINT    — which session this event belongs to
event_count BIGINT    — position of this event within the session
last_event  STRING    — the event type just processed
```

2.2 How it works end-to-end

```
Kafka (user_events)
        │
        ▼
  ┌─────────────┐
  │ user_events │   Flink SQL source table (JSON / latest-offset)
  └─────────────┘
        │
        ▼
  UserEventEnricher(input => TABLE user_events PARTITION BY user_id)
        │
        │  Per-user stateful enrichment:
        │    • login → new session, reset count
        │    • any event → increment count, record last_event
```

```
        │
        ▼
┌─────────────────┐
│ enriched_events │   Flink SQL sink table → Kafka (enriched_events)
└─────────────────┘
```

2.3 Key concepts illustrated

- `ProcessTableFunction` — the Flink 2.x API for stateful, set-semantic UDFs callable from SQL.
- `@StateHint` — declares a POJO whose fields Flink automatically persists per partition key, eliminating manual `ValueState` / `MapState` boilerplate.
- `@ArgumentHint(ArgumentTrait.SET_SEMANTIC_TABLE)` — tells Flink the input is a keyed, stateful virtual processor (set semantics), not a simple row-at-a-time scalar function.
- `PARTITION BY` — the SQL-side mechanism that keys the input table so each `user_id` gets its own isolated state instance.

3.0 Resources

- Create a User-Defined Function with Confluent Cloud for Apache Flink